

NATIONAL INSTITUTE OF TRANSPORT

FACULTY OF INFORMATION TECHNOLOGY AND EDUCATION

DEPARTMEMT OF COMPUTING AND COMMUNICATION TECHNOLOGY

PROJECT PROPOSAL

Project Proposal Title:	**DEVELOPMENT OF LOGISTICS ANALYSIS TOOL FOR FUEL PRICES PREDICTION.**
Project Type:	**DATA SCIENCE AND AI/ML PROJECT**
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Registration Number:	**NIT/BCS/2023/521**
Program:	**BACHELOR'S DEGREE IN COMPUTER SCIENCE**
Level:	**LEVEL 8**
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PROJECT BACKGROUND

Fuel prices play a critical role in deciding the efficiency and sustainability of logistics operations globally, In Tanzania and across many developing economies, fluctuations in fuel costs directly influence transportation expenses, supply chain reliability and uncertainty in budgeting, there is a need for predicting upcoming fuel prices for effective planning.

PROBLEM STATEMENT

Currently, many fuel station operators and logistics companies rely on manual estimations to manage fuel-related costs. This lack of predictive capability leads to poor budgeting, inefficient route planning, and reduced competitiveness.

OBJECTIVES

Main Objective:
To Develop a web-based logistics analysis tool that uses fuel prices data to make analysis, data visualization and future fuel price prediction through machine learning models.

Specific Objectives:
Data collection and cleaning

Data analysis and visualization

Development of prediction model based on fuel prices

Design a user-friendly web-based interface.

Decision making insights according to prediction results.

LITERATURE REVIEW

1. Introduction

In Tanzania, several researches have been done on prediction of petrol and diesel prices by using different machine learning models especially ARIMA model, globally systems have been made to implement the prediction process through secondary data.

This literature review discusses existing researches and systems related to the project and identifies gaps that this project aims to address.

2. Review of Existing Studies and Systems

Ntare p.c(2023) conducted a scientific research on developing fuel prices prediction model where by his findings concluded that ARIMA(p,d,q) was the best model in prediction by using secondary data(2012-2022) due its accuracy.

Gasper Laban and Mbwambo Haika(2023) also conducted a research on determining the problem of predicting crude oil prices in a situation of unusual circumstances, where they concluded that the model must be updated to accurately predict the prices in exceptional circumstances

Also Swoyam Siddharth Nayak(2023) developed the oil and gas prices prediction system in US dollars, The project focused on creating models to predict the closing prices of these fuels based on their opening, highest, and lowest prices then designed a html pages for user interaction to the system.

3. Identified gaps in existing researches and systems;

From the local reviewed literature, it shows that in Tanzania there are only prediction models but no complete systems have been officially published.

All prediction models based on ARIMA model which is less flexible for irregular or missing data.

Secondary data used are not recently (2022)

From The global reviewed literature, the systems have already been made but price prediction bases on USD currency.

4. Relation to the proposed project;

The proposed logistics data analysis tool aims to address the identified gaps by providing;

User friendly system based in Tanzania with easy-to-use features.

Secondary data that are updated (2015 - 2025)

Prediction model based on Prophet Time-series forecasting Model which is more accurate for business forecasting.

Price prediction in Tanzania Shillings (Tsh)

5. Conclusion

By applying these additional features to my project, the logistics analysis system will be improved in terms of price prediction accuracy, user friendly interface, reusability, and local based price predictions.

SIGNIFICANCE OF THE PROJECT

- Provides operational efficiency in fuel and logistics companies
- Enables Risk mitigation by preparing users for fuel prices fluctuation
- Enhance data driven decision making
- Helps in sustainability and resources optimization
- Facilitates technological advancement

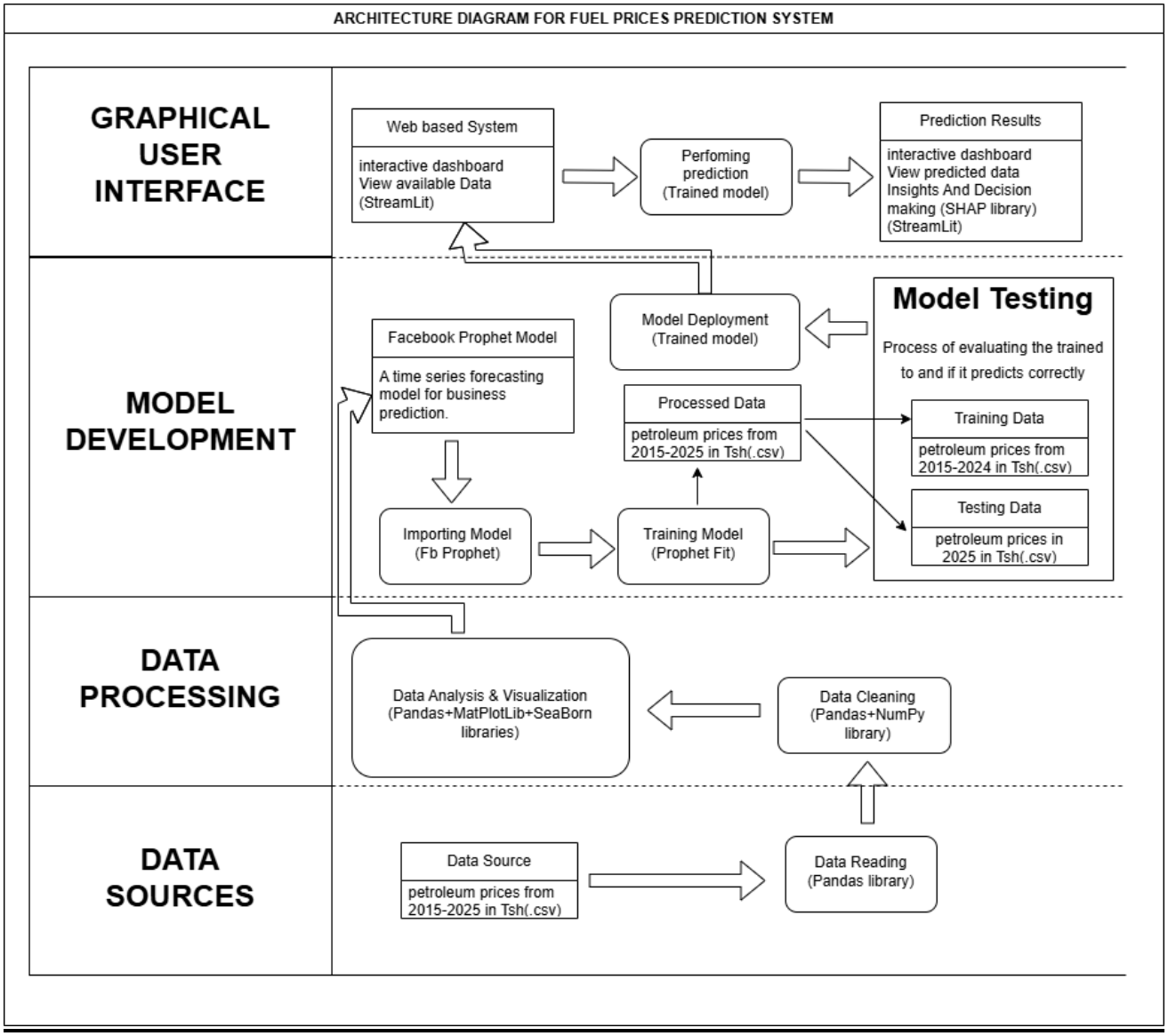
SCOPE OF THE PROJECT

This project will focus on fuel price analysis and prediction, and accurate predictions will be from short to medium terms, (one month to four months) only.

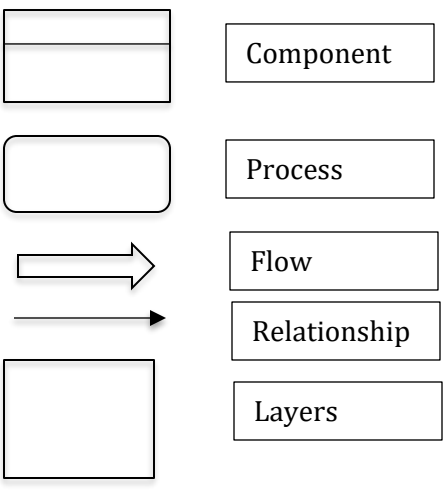
It will not deal with other logistics functions such as routes and it will not cover long term predictions like semi-annually or annually.

SYSTEM ARCHITECTURE DIAGRAM

The following architecture diagram aims to visually represent a system's structure, components, and relationships. It contains Date flow, model training, deployment structure, and system components.

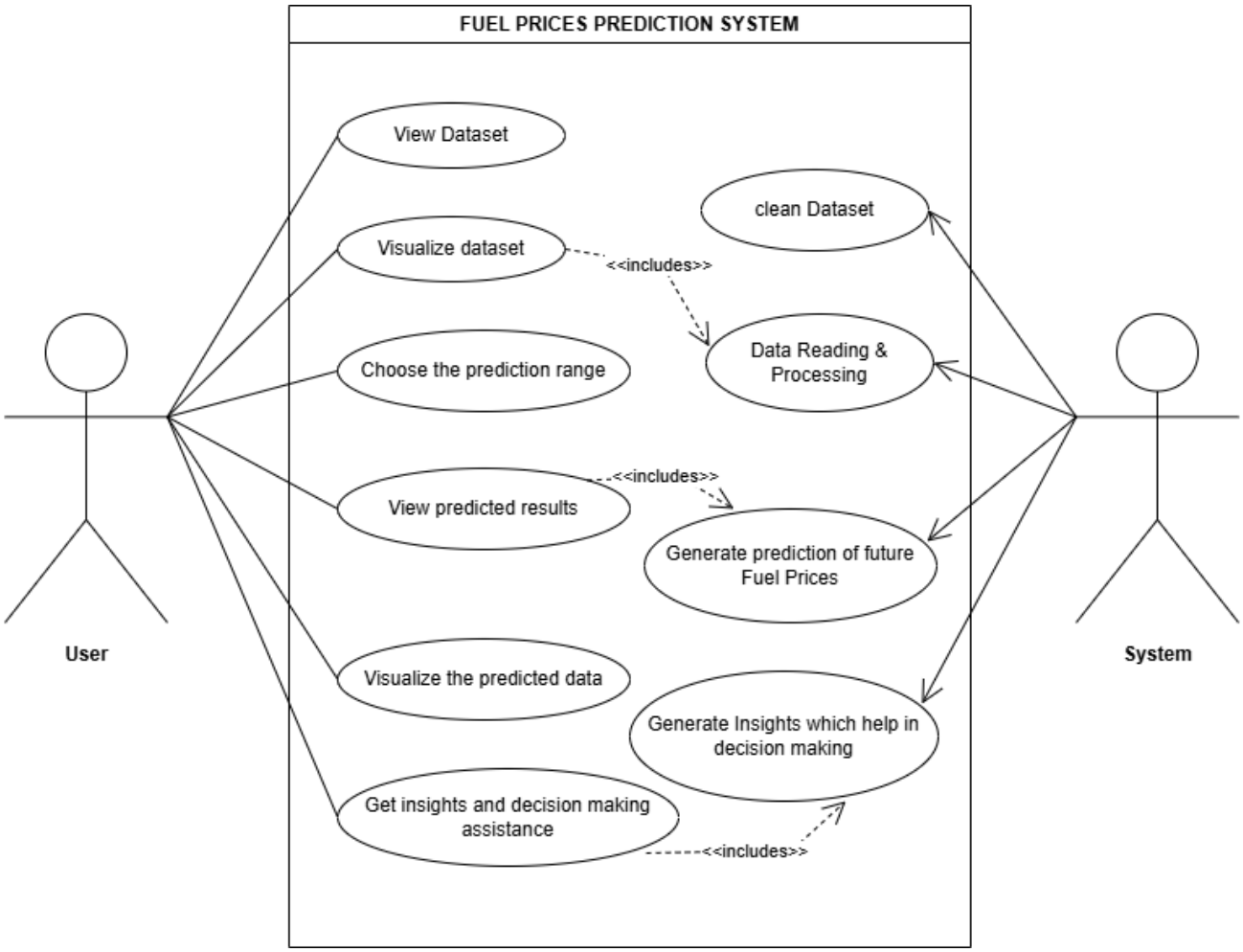


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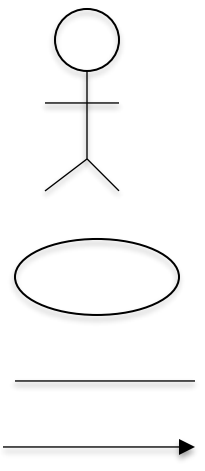


USE CASE DIAGRAM.

The bellow use case diagram aims to show graphical depiction of all system users' interaction with a system and their roles, on other side it shows what functions the system will provide.



KEY;



Actor

Action

Relationship

DirectRelationship

REFERENCES

Gasper, L., & Mbwambo, H. (2023). Forecasting crude oil prices by using ARIMA model: Evidence from Tanzania. Journal of Accounting, Finance and Auditing Studies, 9(2), 158–175.

Swoyam, S. N. (2023). Oil and Gas Price Analysis and Prediction

P. C. Ntare. (2023). Prediction of Petrol and Diesel Prices in Dar Es Salaam using ARIMA models.

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